Chapter 29
Standard Market Models for Rate Derivatives

The interest rate options that trade in the largest volume (caps, floors, and swaptions) are priced by the market using a small family of formulas that are, in essence, Black's model applied three times over. All three trade over the counter, dealer to client under ISDA documentation, in the market Section 10.10 described; none is exchange-listed in meaningful volume. This chapter sets out those formulas, the conventions attached to them, and the two respects in which the standard apparatus has had to change: once because rates went negative, and once because the reference rate is no longer known at the start of the period it applies to.

The standard market models do not describe how interest rates behave. No one believes a single forward rate is lognormal with constant volatility, and Chapters 31 and 32 develop models that take term structure dynamics seriously. What the standard market models provide is a *quotation convention*: a one-to-one map between a price and a number called a volatility, agreed by everyone, so that traders can talk about relative value without arguing about models.

When a trader says three-month expiration swaptions are trading at a normal volatility of 90 basis points, they are not asserting that rates are normally distributed. They are quoting a price in units everyone can invert. The model is a language, and its defects matter only where they make the language ambiguous—which, as Section 29.8 explains, is precisely what happened when rates fell below zero.

The chapter runs from the formula to the surface the market quotes on. Sections 29.1 to 29.4 are Black's model and the instruments it prices: bond options, caps and floors, and the backward-looking caplets the transition created. Sections 29.5 to 29.7 are the conventions attached to them: flat against spot volatilities, European swaptions, and the correlation that makes a cap and a swaption different trades. Sections 29.8 and 29.9 are what negative rates did to the quoting convention, and the volatility cube the market now quotes on; Section 29.10 is how the books are hedged.

29.1 BLACK'S MODEL

Chapter 16 introduced Black's model for options on futures. Its natural home turns out to be interest rate derivatives, for a reason Section 28.6 made precise: each of the products in this chapter has a *natural numeraire* under which the relevant forward rate is a martingale, and under that measure the forward rate can be modeled directly without reference to how it got there.

In its general form, consider a payoff at time T equal to

$$\max(V_T - K, 0)$$

where V_T is the value at T of some variable whose forward value today is F_0. At T the forward and the spot coincide, $F_T = V_T$, so the payoff is equally

$$\max(F_T - K, 0)$$

which is what allows the forward, a martingale under the measure that goes with the product's own numeraire, to be modeled in place of the spot. If the forward is lognormal with volatility σ, the value today is

$$\boxed{c = P(0,T)\,[F_0\,N(d_1) - K\,N(d_2)]} \tag{29.1}$$

with

$$d_1 = \frac{\ln(F_0/K) + \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{29.2}$$

and the corresponding put

$$p = P(0,T)\,[K\,N(-d_2) - F_0\,N(-d_1)] \tag{29.3}$$

Three features make (29.1) the workhorse of this chapter. The discount factor $P(0,T)$ appears *outside* the expectation, so discounting and expectation separate cleanly. The drift of the underlying never appears—only its forward value, which is observable. And σ is the volatility of the *forward* value, not of the spot. The sections that follow identify, for each product, what plays the role of F_0, what plays the role of the discount factor, and over what horizon σ is measured.

29.2 BOND OPTIONS

A bond option confers the right to buy or sell a bond at a specified price on a specified date. Standalone bond options trade, but the economically important ones are embedded: the issuer's right to call a bond before maturity, or the holder's right to put it back.

To apply (29.1), take V_T to be the bond price at option maturity T and F_0 the forward bond price:

$$F_0 = \frac{B_0 - I}{P(0,T)} \tag{29.4}$$

where B_0 is the bond's cash price today and I the present value of coupons paid before T. The volatility σ is that of the forward bond price. Example 29.1 works one through.

Yield volatilities. Bond price volatilities are inconvenient to think about, because a bond's price volatility falls mechanically as it approaches maturity. Traders therefore quote *yield* volatilities, converted using duration. From Section 4.11, $\Delta B \approx -BD^*\Delta y$. Both are proportional volatilities, so put the relation in proportional form:

$$\frac{\Delta B}{B} \;\approx\; -D^* y\,\frac{\Delta y}{y}$$

The price moves by D^*y times the yield's proportional move, and taking standard deviations,

$$\sigma \;\approx\; D^*\,y_0\,\sigma_y \tag{29.5}$$

where σ_y is the forward yield's proportional volatility and D^* the modified duration of the forward bond. A 5% yield with 15% yield volatility on a bond of modified duration 4 implies a price volatility of $4 \times 0.05 \times 0.15 = 3\%$. Example 29.2 shows what skipping the conversion costs.

Example 29.1 A European bond option

A one-year European call struck at 100 is written on a bond whose cash price today is 102.50. The bond pays coupons of 3 at three months and nine months. The relevant discount factors are $P(0,0.25) = 0.9901$, $P(0,0.75) = 0.9704$, and $P(0,1) = 0.9608$. The forward bond price volatility is 4%.

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The present value of the coupons paid before option maturity is

$$I = 3 \times 0.9901 + 3 \times 0.9704 = 5.8815$$

so from (29.4) the forward bond price is

$$F_0 = \frac{102.50 - 5.8815}{0.9608} = 100.5605$$

Then

$$d_1 = \frac{\ln(100.5605/100) + 0.04^2 \times 1/2}{0.04\sqrt{1}} = 0.15973, \qquad d_2 = 0.15973 - 0.04 = 0.11973$$

giving $N(d_1) = 0.563452$ and $N(d_2) = 0.547650$, so from (29.1)

$$c = 0.9608\,[100.5605 \times 0.563452 - 100 \times 0.547650] = 1.822$$

or \$1.82 per 100 of face value. The coupons are subtracted because the option holder does not receive them.

Example 29.2 A quoted yield volatility, converted

Consider a two-year European call on a bond, struck at 98.00, when the forward bond price is 98.50 and $P(0,2) = 0.9124$. The screen quotes a *yield* volatility of 16%; the forward yield is 4.20% and the modified duration of the forward bond is 6.4.

Convert first. From (29.5),

$$\sigma \approx D^* y_0 \sigma_y = 6.4 \times 0.0420 \times 0.16 = 4.3008\%$$

so the total volatility to expiration is $0.043008\sqrt{2} = 0.06082$. Then

$$d_1 = \frac{\ln(98.50/98.00) + 0.043008^2 \times 2/2}{0.043008\sqrt{2}} = 0.11408, \qquad d_2 = 0.11408 - 0.06082 = 0.05326$$

giving $N(d_1) = 0.54541$ and $N(d_2) = 0.52124$, so from (29.1)

$$c = 0.9124\,[98.50 \times 0.54541 - 98.00 \times 0.52124] = 2.41$$

Now price the same option with 16% substituted directly into Black's formula, as though a yield volatility were a price volatility: the answer is **8.31**, three and a half times too large.

The conversion is not a refinement. A bond's price volatility and its yield volatility differ by the factor $D^* y_0$, which here is 0.269, and that factor is what the duration is doing in (29.5).

Callable and puttable bonds. A callable bond is an ordinary bond less a call option held by the issuer; a puttable bond is an ordinary bond plus a put held by the investor. Both can be valued by pricing the straight bond and the embedded option separately. The complication is that the embedded options are typically *Bermudan* (exercisable on any of several dates), which puts them beyond Black's model and into the tree methods of Chapter 31 and the simulation methods of Chapter 32. Black's model remains useful for a first approximation and for quoting.

Option-adjusted spread. Comparing callable bonds by yield is meaningless, because a high yield may reflect a wide credit spread or a valuable call the investor has written. The market compares them by the **option-adjusted spread** (OAS): the constant spread which, added to the benchmark discount curve, makes a model's price for the callable bond equal its market price. It depends on the volatility the model assumes (a higher volatility gives a larger option value and a smaller OAS), so an OAS quoted without its volatility is not a well-defined number; Section 31.11 computes one.

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29.3 INTEREST RATE CAPS AND FLOORS

An **interest rate cap** protects a floating-rate borrower against rates rising above a level K. It is written on a notional L over a sequence of accrual periods, and pays, for the period $[T_i, T_{i+1}]$ with accrual factor τ_i, at the period's end T_{i+1},

$$L\,\tau_i \max(R_i - K, 0) \tag{29.6}$$

where R_i is the reference rate for that period. A **floor** is the mirror image, paying $L\tau_i \max(K - R_i, 0)$.

The option covering one period is a **caplet**, or for a floor a floorlet, and the essential observation is that each one settles on its own period's rate, whatever the others do. A cap is a *portfolio* of caplets, not a compound option, so it can be valued one period at a time and summed.

Applying Black's model to a single caplet, with F_i the forward rate for $[T_i, T_{i+1}]$:

$$\text{caplet}_i = L\,\tau_i\,P(0,T_{i+1})\,[F_i\,N(d_1) - K\,N(d_2)] \tag{29.7}$$

with d_1, d_2 from (29.2) using the volatility σ_i of that forward rate over $[0, T_i]$: the rate fixes at T_i, so that is when the option's uncertainty is resolved, even though the payment does not arrive until T_{i+1}. Discounting is done with $P(0, T_{i+1})$, the bond maturing on the payment date, which, as Section 28.6 established, is exactly the numeraire under which F_i is a martingale, so no adjustment is required.

Caplets and bond options. A caplet is also a bond option, and the equivalence is an identity rather than an approximation. The reference rate is read off the bond price prevailing at T_i,

$$R_i = \frac{1}{\tau_i}\left(\frac{1}{P(T_i,T_{i+1})} - 1\right)$$

and the payoff (29.6) is already known at T_i, so discounting it back over the accrual period multiplies it by $P(T_i, T_{i+1})$ and the bond prices cancel:

$$L\,\tau_i\,P(T_i,T_{i+1})\max(R_i - K, 0) = L\,(1+\tau_i K)\max\left(\frac{1}{1+\tau_i K} - P(T_i,T_{i+1}),\ 0\right)$$

A caplet is therefore $L(1+\tau_i K)$ European *puts* on the T_{i+1} zero-coupon bond, expiring at T_i and struck at $1/(1+\tau_i K)$. They are puts because the bond price falls when the rate rises. Nothing has been assumed about how the rate is distributed (the identity is between contracts, not between models), and it is what allows a model of the short rate to price a cap in closed form, since such a model delivers the distribution of $P(T_i, T_{i+1})$ directly. Section 31.7 does that.

Example 29.3 Valuing a caplet

A caplet covers the three-month period beginning in one year, on a notional of \$10 million. The forward rate for that period is $F = 4.00\%$, the strike is $K = 4.25\%$, the volatility is $\sigma = 20\%$, the accrual factor is $\tau = 0.25$, and $P(0, 1.25) = 0.9520$.

Taking the horizon as $T = 1$ (the rate is fixed at the start of the period),

$$d_1 = \frac{\ln(0.04/0.0425) + 0.20^2 \times 1/2}{0.20\sqrt{1}} = -0.2031, \qquad d_2 = -0.2031 - 0.20 = -0.4031$$

so $N(d_1) = 0.41952$ and $N(d_2) = 0.34343$. From (29.7),

$$\text{caplet} = 10{,}000{,}000 \times 0.25 \times 0.9520\,[0.04 \times 0.41952 - 0.0425 \times 0.34343] = \$5{,}200$$

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Cap–floor parity. A useful identity follows from the payoffs: since $\max(R-K,0)-\max(K-R,0)=R-K$,

$$\text{cap} - \text{floor} = \text{payer swap struck at } K \tag{29.8}$$

A long cap and short floor at the same strike replicate a swap paying fixed at K and receiving floating. This is a model-free relationship, and it provides an immediate consistency check: cap and floor quotes that violate it are arbitrageable regardless of anyone's view on volatility.

Who buys a cap. The natural buyer is a borrower with floating-rate debt: a company or property owner whose loan resets quarterly buys a cap to hold its funding cost at the strike while keeping the benefit of lower rates, and lenders often require it.

Who buys a floor, and the collar. A floor is bought by whoever loses when rates fall: a bank with floating-rate assets, an investor in floating-rate notes, a lender whose loan carries no floor of its own. A borrower can pay for a cap by selling a floor, which turns the hedge into an **interest rate collar**: the funding cost is held between the two strikes, and the floor strike is often set so the two premiums offset, a zero-cost collar. By cap–floor parity (29.8), a collar whose two strikes coincide is a payer swap.

Listed rate options. Caps and floors are over-the-counter. The exchange-traded counterparts are options on SOFR futures (including mid-curve options, which expire before the underlying futures contract) and on Treasury futures. They are American-style, but the early-exercise premium is small at the expirations where they trade, so they are priced on the futures price with Black's model or, as the market usually quotes them, with Bachelier's model in normal volatility (Section 29.8), and need no convexity adjustment, because the futures price is already a martingale under the risk-neutral measure (Section 28.7). Asset managers, hedge funds and bank treasuries use them to hold rate optionality without a bilateral contract.

Which caplets are included. Under LIBOR the first caplet of a **spot-starting** cap (one whose first accrual period begins today) was excluded from the quote, because the first period's rate was already fixed and the caplet's payoff therefore known. That convention no longer makes sense, for the reason developed next.

29.4 BACKWARD-LOOKING CAPLETS

Section 29.3 quietly assumed the reference rate is fixed at the start of its accrual period, so that a caplet is an option expiring at T_i on a quantity known at T_i. Chapter 4 established that this is no longer how the reference rate works. A compounded risk-free rate accrues across $[T_i, T_{i+1}]$ and is not known until the period ends. An option on such a rate is a **backward-looking** caplet, and it differs from its forward-looking predecessor in a way that is small in magnitude but conceptually important.

Nothing is exercised by decision here. A caplet settles automatically against the realized rate, so its "expiration" is the date the payoff becomes *known*, not a date on which somebody chooses: T_i when the rate is fixed in advance, T_{i+1} when it is compounded across the period.

How much uncertainty is there?

Consider the rate for the period $[T_1, T_2]$, with $\tau = T_2 - T_1$, and write σ for its volatility, in the sense of Section 29.1: the volatility of a forward value, taken as constant. Before T_1 no fixing has entered it yet, so it accumulates uncertainty as a forward-looking rate does: variance $\sigma^2 t$ at time t. Its forward is the same F_0, as the pricing below shows. The two rates part company only once compounding starts.

From T_1, each day's overnight fixing that has already been observed is locked in. As the period elapses, a growing fraction of the final compounded rate is already determined, so the

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remaining sensitivity to rate movements declines, linearly to a good approximation, reaching zero at T_2. This is the same decaying-delta effect that Section 30.3 notes for averaging futures.

Modeling that decay as linear, the rate's volatility at a time u inside the period is $\sigma(T_2-u)/\tau$, falling from σ at T_1 to zero at T_2, and the variance accumulated between T_1 and T_2 is

$$\int_{T_1}^{T_2} \sigma^2 \left(\frac{T_2-u}{\tau}\right)^2 du \;=\; \frac{\sigma^2\tau}{3} \tag{29.9}$$

Adding the variance accumulated before T_1 gives the total:

$$\text{Var} = \sigma^2\left(T_1 + \frac{\tau}{3}\right) \tag{29.10}$$

Figure 29.1 shows the two paths, drawn with $\tau = T_1$ so the effect is visible. The forward-looking rate stops accumulating uncertainty at T_1; the backward-looking rate continues past it, though ever more slowly as more of the average is locked in, adding a further $\tau/3$ years' worth of variance.

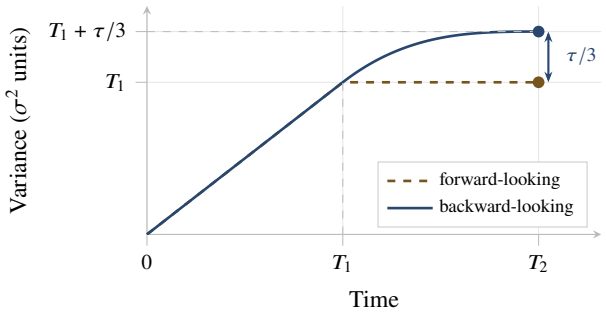


Figure 29.1: Uncertainty accumulated by a forward-looking and a backward-looking rate, drawn with $\tau = T_1$.

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The measure does not change: a caplet pays at T_2 either way, so $P(0,T_2)$ is the numeraire in both cases. What changes is the process modeled under it. Section 29.3 used the forward rate $F(t)$ for $[T_1,T_2]$, which stops moving at T_1, and a rate still accumulating at T_2 needs an extension of it.

The extension comes from a trading strategy: buy the T_1 bond and, at T_1, roll the proceeds at the overnight rate through to T_2. Its value $C(t)$ is a traded portfolio throughout, so $C(t)/P(t,T_2)$ is a martingale under the T_2-forward measure, and therefore so is

$$R(t) = \frac{1}{\tau}\left(\frac{C(t)}{P(t,T_2)} - 1\right) \tag{29.11}$$

Before T_1, $C(t) = P(t,T_1)$ and $R(t)$ is the old forward rate unchanged; at T_2 it is the compounded rate the caplet settles against; and $R(0) = F_0$, so both caplets are priced off the same forward. Only the total variance changes: the martingale now runs to T_2 instead of freezing at T_1, and (29.10) is what it accumulates. So price a backward-looking caplet with Black's formula (29.7), but replace $\sigma^2 T_1$ with $\sigma^2(T_1+\tau/3)$ wherever it appears. Equivalently, use an effective volatility

$$\sigma_{\text{eff}} = \sigma\sqrt{\frac{T_1+\tau/3}{T_1}} \tag{29.12}$$

over the horizon T_1. Example 29.4 applies it to the caplet of Example 29.3.

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Example 29.4 The same caplet, backward-looking

Take Example 29.3 and treat the rate as compounded in arrears rather than fixed in advance. Here $T_1 = 1$ and $\tau = 0.25$, so

$$\sigma^2\left(T_1 + \frac{\tau}{3}\right) = 0.20^2\left(1 + \frac{0.25}{3}\right) = 0.043333, \qquad \sqrt{0.043333} = 0.20817$$

Repeating the calculation with $\sigma\sqrt{T}$ replaced by 0.20817:

$$d_1 = \frac{\ln(0.04/0.0425) + 0.043333/2}{0.20817} = -0.1871, \qquad d_2 = -0.1871 - 0.20817 = -0.3953$$

giving $N(d_1) = 0.425772$, $N(d_2) = 0.346305$, and

$$\text{caplet} = 10{,}000{,}000 \times 0.25 \times 0.9520\,[0.04 \times 0.425772 - 0.0425 \times 0.346305] = \$5{,}505$$

against \$5,200 for the forward-looking version—about **5.9% more valuable**.

Two observations follow.

Backward-looking caplets are worth more. Since $T_1 + \tau/3 > T_1$ always, the option on a compounded rate is the more valuable. It is exposed to rate movements for longer.

The first caplet is no longer trivial. Under LIBOR the first caplet of a spot-starting cap had $T_1 = 0$ and hence zero variance: the rate was already fixed, and the caplet was a known cash flow, conventionally excluded from the quote. Under a compounded rate, $T_1 = 0$ gives variance $\sigma^2\tau/3$, which is not zero. That caplet now carries genuine optionality, and the old convention of dropping it silently understates the price.

The effect scales with the accrual length. For quarterly periods $\tau/3$ is a month, negligible against a five-year expiration but material against a three-month one. For annual periods, or for the front of the curve, it matters.

29.5 FLAT AND SPOT VOLATILITIES

A cap is a portfolio of caplets, each with its own volatility. The market, however, quotes a single number per cap. This creates two distinct volatility concepts that are easily confused and must be kept apart.

- A **flat volatility** is the single volatility which, applied to *every* caplet in a given cap, reproduces the cap's market price. It is a quotation device, not a property of any individual caplet.
- A **spot volatility** (also called a caplet or forward volatility) is the volatility of one specific caplet.

A flat volatility is thus a kind of average of the spot volatilities of the caplets making up that cap. The two coincide only for the shortest cap, the one containing a single caplet. That averaging is visible in Figure 29.2: because a flat volatility summarizes every caplet up to its maturity, its curve peaks later, peaks lower, and decays more slowly than the spot curve underneath it.

Extracting spot volatilities from quoted flat volatilities is a bootstrap, exactly parallel to the zero curve bootstrap of Chapter 4. The price of a two-year cap less the price of a one-year cap is the value of the caplets falling in the second year; those caplets' spot volatilities are whatever reproduces that difference, given the spot volatilities already extracted for year one. Example 29.5 carries out one step.

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Example 29.5 Bootstrapping spot volatilities

Annual-reset caps struck at 4.00% are quoted at flat volatilities of 20.0% for the two-year and 21.0% for the three-year. Take the forward rates to be 4.00% for $[1, 2]$ and 4.30% for $[2, 3]$; discount factors are $P(0, 2) = 0.9246$ and $P(0, 3) = 0.8890$. Work per unit of notional.

Two-year cap. It contains a single caplet, covering $[1, 2]$ (the first period is excluded, as under the LIBOR convention of Section 29.3), so its spot volatility is the quoted flat volatility, $\sigma_1 = 20.0\%$. At the money with $T = 1$, $d_1 = 0.20\sqrt{1}/2 = 0.1$ and $d_2 = -0.1$, so

$$\text{caplet}_{[1,2]} = 0.9246 \times 0.04\,(0.539828 - 0.460172) = 0.0029460$$

Three-year cap. Valuing *both* caplets at the quoted flat volatility of 21% gives 0.0030928 for $[1, 2]$ and 0.0058161 for $[2, 3]$, a total cap price of

$$0.0030928 + 0.0058161 = 0.0089089$$

The bootstrap step. The $[2, 3]$ caplet must account for the remainder:

$$0.0089089 - 0.0029460 = 0.0059629$$

Solving Black's formula numerically for the volatility that reproduces this value gives

$$\sigma_2 = 21.7\%$$

So a flat volatility curve of 20.0%, 21.0% corresponds to spot volatilities of 20.0%, 21.7%. The second caplet's own volatility exceeds the three-year cap's quote, because that quote is diluted by the lower-volatility first caplet inside it. Using 21% to value the $[2, 3]$ caplet in isolation would understate it by about 2.5%.

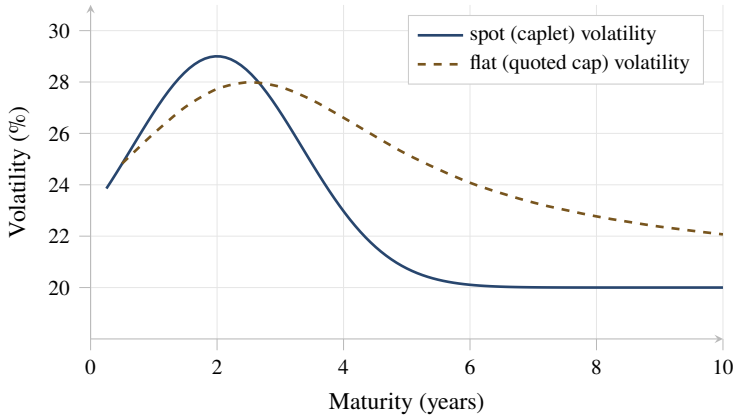


Figure 29.2: Representative shapes of flat and spot volatility curves. Illustrative rather than market data.

The characteristic shape in Figure 29.2, rising to a hump at two to three years and then declining, recurs across currencies; it gives way when the policy setting itself becomes the uncertainty, as in 2022. The usual explanation is that very short rates are anchored by the central bank's current policy setting and very long rates by expectations about the long run, while the intermediate maturities carry most of the uncertainty about the path of policy in between.

A flat volatility quoted for a five-year cap and a flat volatility quoted for a ten-year cap describe overlapping portfolios; the difference between them is not the volatility of anything. Comparing the two directly, or reading a caplet's volatility off them by interpolation, is a common and expensive error. Convert to spot volatilities before doing anything analytical.

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29.6 EUROPEAN SWAPTIONS

A **swaption** is an option to enter an interest rate swap. A **payer** swaption confers the right to pay fixed and receive floating; a **receiver** swaption the right to receive fixed. They are described by two maturities (the option expiration and the underlying swap's tenor), so a "5×10" is a five-year option on a ten-year swap.

The key to valuing them is to recognize what the option is really on. At expiration T, the holder of a payer swaption struck at K exercises if the prevailing swap rate S_T exceeds K, gaining $L\tau_i(S_T - K)$ on each of the n payment dates. The total value at exercise is

$$L \sum_{i=1}^{n} \tau_i P(T, t_i) \max(S_T - K, 0) = L\, A(T) \max(S_T - K, 0)$$

where $T < t_1 < \cdots < t_n$ are the payment dates of the fixed leg of the swap the option delivers, τ_i the accrual factor for each, and $A(T)$ the annuity of Section 28.8, the present value at T of one unit paid on each of those dates. So a swaption is an option on the swap rate, with a payoff multiplied by the annuity. In the major currencies that fixed leg now pays annually (Section 7.1), so τ_i is about one and n is the swap's tenor in years, which is what Example 29.6 assumes. Legacy dollar swaps paid fixed semiannually against quarterly floating, and some currencies still do.

That structure points to the right numeraire. Taking the annuity as numeraire (Section 28.8), the swap rate is a martingale, and Black's model applies with the annuity replacing the discount factor:

$$\text{payer} = L\, A(0)\, [S_0\, N(d_1) - K\, N(d_2)] \tag{29.13}$$

$$\text{receiver} = L\, A(0)\, [K\, N(-d_2) - S_0\, N(-d_1)] \tag{29.14}$$

with d_1, d_2 from (29.2) using the forward swap rate S_0 and its volatility, and

$$A(0) = \sum_{i=1}^{n} \tau_i\, P(0, t_i) \tag{29.15}$$

Chapter 30 exploits the annuity's role as numeraire directly: the CMS convexity adjustment arises precisely because a CMS coupon is paid at a single date rather than across the annuity.

Example 29.6 Valuing an at-the-money swaption

Consider a five-year option on a five-year annual-pay swap, notional \$100 million. The forward swap rate is $S_0 = 4.00\%$, the strike is $K = 4.00\%$, and the volatility is $\sigma = 20\%$.

With a flat 4% curve, the discount factors for the five payment dates (years 6 through 10) are 0.790315, 0.759918, 0.730690, 0.702587, and 0.675564, so from (29.15)

$$A(0) = 3.659074$$

At the money, $d_1 = 0.20\sqrt{5}/2 = 0.22361$ and $d_2 = -0.22361$, giving $N(d_1) = 0.58847$ and $N(d_2) = 0.41153$. From (29.13),

$$\text{payer} = 100{,}000{,}000 \times 3.659074 \times 0.04 \times (0.58847 - 0.41153) = \$2{,}589{,}700$$

or about 2.59% of notional. At the money the payer and receiver have equal value, as put–call parity requires: their difference is a forward swap struck at the forward rate, which is worth zero.

Swaptions and bond options. A payer swaption is equivalent to a put option on a coupon bond: exercising the right to pay fixed at K is exercising the right to sell, at par, a bond paying coupon K. Receiver swaptions correspond to call options on the same bond. The swap-rate formulation is nonetheless the market standard, because a swap rate's volatility, unlike a bond price's, does not shrink mechanically as the underlying approaches maturity.

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Who uses swaptions. A borrower planning to issue fixed-rate debt buys a payer swaption to cap the rate it will pay. Mortgage servicers and insurers with minimum-rate guarantees buy receiver swaptions, because falling rates bring prepayments or deepen the guarantees. Asset managers and pension funds sell swaptions to earn the premium, which is why the market's volatility is not set by hedgers alone. Issuers of callable debt sell dealers the call they hold, as a receiver swaption, which supplies long-dated volatility while mortgage hedgers demand it; the balance of those flows, more than any model, sets the level of long-dated swaption volatility.

29.7 CAPS VERSUS SWAPTIONS: THE ROLE OF CORRELATION

Caps and swaptions written on the same schedule and strike look superficially like the same trade. They are not.

A cap pays $\sum_i \tau_i \max(F_i - K, 0)$: it is a **portfolio of options**, one on each forward rate, exercised independently. A payer swaption pays $A \max(S - K, 0)$, and since the swap rate is approximately a weighted average of the same forward rates, it is an **option on a portfolio**.

The distinction has a definite direction. For any nonnegative weights w_i summing to one,

$$\sum_i w_i \max(F_i - K, 0) \;\geq\; \max\left(\sum_i w_i F_i - K,\; 0\right) \tag{29.16}$$

because a sum of positive parts is at least the positive part of the sum. Hence, since each caplet is worth at least its intrinsic value at the swaption's expiration, **a cap starting then is always worth at least as much as the corresponding swaption**.

The intuition is diversification, running backward. The cap holder collects on whichever rates happen to exceed the strike, and is not penalized for those that do not. The swaption holder owns one option on the average, so a rate that finishes far above the strike is partly canceled by another that finishes below it.

That tells us what governs the size of the gap: the **correlation** between forward rates. If all forward rates started level and moved in perfect lockstep, they would always finish on the same side of the strike, no cancellation could occur, and (29.16) would hold with equality. The less correlated they are, the more the averaging suppresses the swap rate's variance, and the cheaper the swaption becomes relative to the cap.

This can be made quantitative. Suppose n forward rates each have volatility σ, enter the swap rate with equal weight, and have uniform pairwise correlation ρ. The variance of the average is then $\sigma^2 \left[1/n + (1 - 1/n)\,\rho\right]$, so the swap rate's volatility is

$$\sigma_{\text{swap}} = \sigma\sqrt{\rho + \frac{1-\rho}{n}} \tag{29.17}$$

At $\rho = 1$ this returns σ, and the two products carry the same volatility. At $\rho = 0$ it falls to $\sigma/\sqrt{n}$. For a ten-period swap with $\rho = 0.7$, the swaption's volatility is $\sigma\sqrt{0.73} = 0.854\,\sigma$—some 15% below the caplet volatility, purely from imperfect correlation.

The argument also runs in reverse. Correlations between forward rates are not directly observable, but cap and swaption prices are. The ratio between them is therefore the market's main source of information about forward-rate correlation, and it is the primary calibration input for the multi-factor term structure models of Chapter 32. A model calibrated to caps alone will price swaptions badly, because caps contain almost no information about correlation: each caplet depends on one rate at a time.

29.8 NORMAL AND SHIFTED LOGNORMAL MODELS

Every formula so far assumes the underlying rate is lognormal. That assumption carries an implication which went unexamined for decades: a lognormal variable cannot be negative or

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zero. When European rates fell below zero in the mid-2010s, Black's model did not merely become inaccurate—it became *undefined*. A negative forward rate cannot be substituted into $\ln(F_0/K)$.

Case Study 29.1 When Black's model broke

The European Central Bank moved its deposit facility rate below zero in June 2014. Within a year three-month Euribor had fixed below zero for the first time, in April 2015, and short euro swap rates had followed; by 2016 substantial portions of the euro and Swiss curves traded at negative yields. The Bank of Japan followed in January 2016.

For the swaption market this was not an academic inconvenience. Brokers quote volatility, not price, and the standard quotation convention had suddenly stopped being able to represent the market. Screens showed blanks. Risk systems that inverted Black's formula to imply a volatility returned errors on live positions.

The market responded in two ways within a matter of months. Some desks adopted a **shifted lognormal** convention, adding a fixed displacement (typically 1% to 3%) to both rate and strike so that the shifted quantities stayed positive and Black's formula could be applied unchanged. Others moved to quoting **normal** volatilities under the Bachelier model, which places no restriction on sign. Normal volatility eventually became the euro and yen market standard and has since been widely adopted for dollar swaptions as well.

The episode is a useful reminder that a modeling assumption can survive for decades not because it is right but because it has never been tested where it fails.

The normal (Bachelier) model

Assume instead that the rate is normally distributed, with $dF = \sigma_N \, dW$ for a constant **normal volatility** σ_N quoted in basis points per annum rather than as a percentage. The value of a payer-type option is

$$c = P(0,T)\left[(F_0 - K)\,N(d) + \sigma_N\sqrt{T}\,\varphi(d)\right], \qquad d = \frac{F_0 - K}{\sigma_N\sqrt{T}} \tag{29.18}$$

where φ is the standard normal density. The formula is defined for any real F_0 and K, positive or negative.

At the money the expression collapses, since $d = 0$, $N(0) = 1/2$, and $\varphi(0) = 1/\sqrt{2\pi}$:

$$c_{\text{ATM}} = P(0,T)\,\sigma_N\sqrt{\frac{T}{2\pi}} \tag{29.19}$$

An at-the-money option's value is proportional to the normal volatility and to the square root of time. Traders use this constantly for order-of-magnitude checks.

Converting between conventions

Because both conventions are one-to-one maps between price and volatility, a price quoted one way can always be restated the other way. Near the money the relationship is approximately

$$\sigma_N \;\approx\; \sigma_{\text{LN}} \times F_0 \tag{29.20}$$

where σ_{LN} is the lognormal (Black) volatility. This is just the statement that a proportional volatility applied to a level gives an absolute volatility. Example 29.7 tests it on the swaption of Example 29.6.

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Example 29.7 Lognormal and normal volatility for the same option

The swaption of Example 29.6 has an at-the-money value of 2.59% of notional, with $A(0) = 3.659074$. Dividing out the annuity, the option value per unit of annuity is

$$0.04 \times (0.58847 - 0.41153) = 0.0070776$$

Setting this equal to the at-the-money normal value $\sigma_N \sqrt{T/2\pi}$ from (29.19),

$$\sigma_N = \frac{0.0070776\sqrt{2\pi}}{\sqrt{5}} = 0.007934$$

or **79.3 basis points**. The rule of thumb (29.20) predicts $0.20 \times 4.00\% = 80$ basis points—within a basis point, which is why the approximation is used so freely.

Which to use

The two models make genuinely different statements about how rates move. Black's σ is a proportional volatility ($dF = \sigma F\, dW$), so the standard deviation of the rate's change, measured in basis points, scales with the level: at a quoted 20%, a 5% rate has an annual standard deviation of 100 basis points against 50 for a 2.5% rate. Bachelier's σ_N is that standard deviation quoted directly, and is therefore the same number of basis points at any level.

The empirical evidence is regime-dependent, and not in one direction: rate moves look close to absolute, independent of the level, when rates sit in a middle band of about 2% to 6%, and closer to proportional both below that band and above it, the pattern Deguillaume, Rebonato, and Pogudin (2013) found across currencies and decades. Practice has settled on normal volatilities for quotation, with the understanding that the number will need to be re-marked as the level of rates changes.

Figure 29.3 shows the difference where it matters most, for a forward of 0.5%. The lognormal density is confined to positive values and cannot represent a negative outcome at all; the normal density assigns genuine probability to the shaded region below zero.

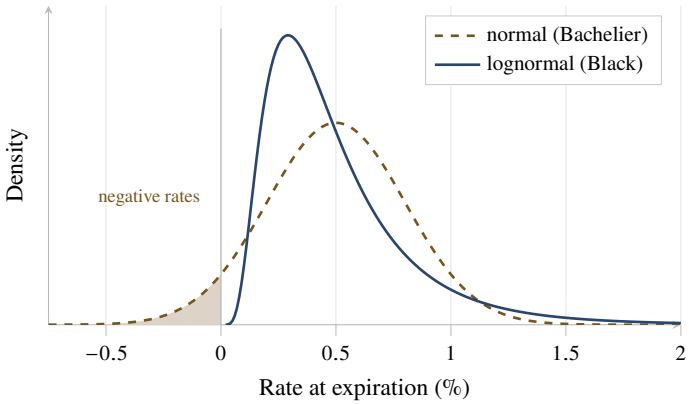


Figure 29.3: Distributions of a rate with a forward of 0.5% under the two conventions.

Shifted lognormal

The shifted lognormal convention is a compromise. Choose a displacement δ large enough that $F_0 + \delta$ and $K + \delta$ are both comfortably positive, and apply Black's formula to the shifted quantities. It preserves the lognormal machinery, permits negative rates down to $-\delta$, and

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interpolates between the two extremes: large δ approaches normal behavior, $\delta = 0$ recovers Black.

Its drawback is that δ is one more convention to agree. A shifted lognormal volatility is meaningless without the shift that goes with it, and quotes carrying different shifts are not comparable. This is the same trap as flat versus spot volatilities, and the same discipline applies: establish the convention before comparing the numbers.

29.9 THE VOLATILITY CUBE

A single volatility number is enough to quote one option. Describing the whole swaption market requires three dimensions:

- **option expiration**, from one month to thirty years;
- **swap tenor**, the length of the underlying swap;
- **strike**, usually expressed as an offset from the at-the-money forward.

The resulting object is the **volatility cube**. Fixing the strike at the money gives the volatility surface over expiration and tenor; fixing expiration and tenor and varying strike gives a smile, of the kind Chapter 19 discussed for equities.

Two regularities stand out. Along the expiration dimension, volatility displays the hump of Figure 29.2. Along the tenor dimension, volatility generally *declines*: a thirty-year swap rate is an average of forward rates over thirty years, and averaging suppresses variance (exactly the mechanism of Section 29.7), so long-tenor swap rates move less than short-tenor ones. Figure 29.4 shows the at-the-money slice of the cube with both effects visible at once: the ridge running along two to three years' expiration, where uncertainty about the policy path is greatest, and the steady decline as swap tenor lengthens.

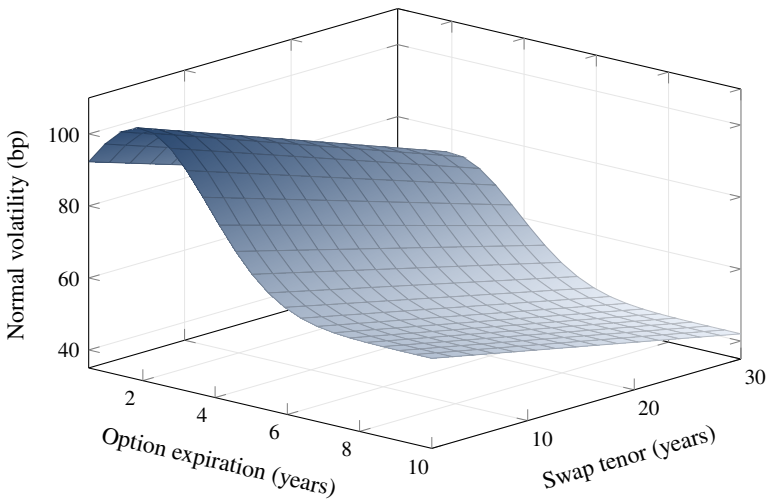


Figure 29.4: Normal volatility against option expiration and swap tenor. Representative rather than market data.

Fitting the cube: SABR

Interpolation within the cube is constrained, because an arbitrary interpolation can produce a surface admitting arbitrage: calendar spreads with negative value, or butterfly spreads implying negative probability density. Practitioners therefore fit a parametric model, and for the strike dimension the standard choice is **SABR**, proposed by Hagan and co-authors in 2002. It is a

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stochastic-volatility model in the sense of Section 21.4, but built around a different requirement from Heston's: it must be fast enough to refit every swaption smile on the grid, many times a day. It posits a forward rate whose own volatility is stochastic:

$$dF = \alpha F^{\beta}\, dW_1, \qquad d\alpha = \nu\, \alpha\, dW_2, \qquad dW_1\, dW_2 = \rho\, dt \tag{29.21}$$

The model has four parameters, each with a distinct job:

- α sets the overall level of volatility, and is what the at-the-money quote pins down.
- β governs the **backbone**, how the at-the-money volatility moves when the forward rate itself moves. At $\beta = 1$ the model is lognormal and at-the-money volatility is unchanged by a shift in F; at $\beta = 0$ it is normal, and lognormal volatility moves inversely with F. This is the same choice as Section 29.8, now made continuously. At $\beta = 0$ the dynamics allow negative rates, though the expansion (29.22) below, written in Black volatility and $\ln(F/K)$, does not; desks then use Hagan's normal-volatility form of it, or the shifted SABR described at the end of this section.
- ρ produces the skew. A negative correlation between rate and volatility tilts the smile downward to the right, which is the shape rate markets normally display.
- ν, the volatility of volatility, produces the curvature, how much the wings lift relative to a straight line through the middle.

That division is the same one Heston makes in Section 21.4. What made SABR the market standard is not the dynamics, which are unremarkable, but the closed-form approximation Hagan derived for the Black (lognormal) implied volatility $\sigma_B(K)$ they produce at strike K and expiration T, with no integral and no simulation:

$$\sigma_B(K) = \frac{\alpha}{(FK)^{(1-\beta)/2}\, D(K)} \cdot \frac{z}{x(z)} \cdot (1 + C\,T) \tag{29.22}$$

where, writing $\ell = \ln(F/K)$,

$$D(K) = 1 + \frac{(1-\beta)^2}{24}\ell^2 + \frac{(1-\beta)^4}{1920}\ell^4, \qquad z = \frac{\nu}{\alpha}(FK)^{(1-\beta)/2}\ell$$

$$x(z) = \ln\left[\frac{\sqrt{1 - 2\rho z + z^2} + z - \rho}{1 - \rho}\right],$$

$$C = \frac{(1-\beta)^2}{24}\frac{\alpha^2}{(FK)^{1-\beta}} + \frac{1}{4}\frac{\rho\beta\nu\alpha}{(FK)^{(1-\beta)/2}} + \frac{2 - 3\rho^2}{24}\nu^2$$

The formula has three factors, and each does one job: the first is the level, the second bends the smile away from it, and the third is the maturity correction. At $K = F$ the middle factor is 1 and the whole thing collapses to $\alpha F^{\beta-1}(1 + CT)$. A single evaluation takes microseconds, which is what meets the speed requirement this subsection opened with.

β is usually fixed rather than fitted. It is nearly redundant with ρ on a single smile, so fitting both produces unstable parameters that jump between days without the market having moved. [TN29] makes the redundancy precise: to first order in log-moneyness the skew depends on β and ρ only through one combination of them, and the curvature is where they separate again.

The calibration recipe. Four parameters cannot be identified from a small number of quotes without structure, so desks proceed in a fixed order:

1. **Fix** β by judgment or by regressing historical at-the-money volatility on the level of the forward, never by fitting it to a single day's smile, for the reason just given. Rate desks commonly use $\beta = 0.5$.
2. **Solve** α **from the at-the-money quote**, given trial values of ρ and ν. This is a cubic in α, and forces the fit through the most liquid point exactly.

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3. **Fit ρ and ν to the wings**, re-solving for α at each step so the at-the-money quote is never violated.

Example 29.8 Fitting SABR to a swaption smile

A five-year option on a five-year swap has a forward swap rate of $F = 4.00\%$. Three strikes are quoted, in lognormal volatility:

$$\sigma_B(3.00\%) = 23.6\%, \qquad \sigma_B(4.00\%) = 20.0\%, \qquad \sigma_B(5.00\%) = 18.3\%$$

Fix $\beta = 0.5$. Three quotes then determine the three remaining parameters, and solving (29.22), rounded, gives

$$\alpha = 0.0387, \qquad \rho = -0.30, \qquad \nu = 0.35$$

Checking the at-the-money point. With $F^{1-\beta} = \sqrt{0.04} = 0.2$, the leading term is $\alpha/0.2 = 0.1935$, and

$$C = \frac{0.25}{24}\frac{0.0387^2}{0.04} + \frac{1}{4}\frac{(-0.30)\times 0.5\times 0.35\times 0.0387}{0.2} + \frac{2-3\times(-0.30)^2}{24}\times 0.35^2 = 0.006681$$

Hence $\sigma_B(F) = 0.1935\,(1 + 0.006681\times 5) = 20.00\%$, matching the quote.

Reproducing the wings. Substituting $K = 3\%$ gives $z = 0.4842$, $x(z) = 0.4403$, hence $z/x(z) = 1.0998$, and (29.22) returns 23.60%. At $K = 5\%$, $z = -0.4268$ and $z/x(z) = 0.9655$, giving 18.26%. Both agree with the quotes to the one decimal place at which they are quoted.

Using the fit. The calibration's value is what it says about strikes no one quoted:

Strike	2.5%	3.0%	3.5%	4.0%	4.5%	5.0%	5.5%
σ_B	26.25%	23.60%	21.54%	20.00%	18.92%	18.26%	17.91%

The smile is steeply downward-sloping near the money and flattens in the high-strike wing—the characteristic shape of a rate market, and in its downward slope much like the equity skew of Chapter 19.

Figure 29.5 plots the result: the at-the-money point is matched by step 2 of the recipe, and the wings are what ρ and ν must reproduce.



Figure 29.5: The SABR fit of Example 29.8, with $\beta = 0.5$, $\alpha = 0.0387$, $\rho = -0.30$, $\nu = 0.35$.

Where the approximation breaks. Equation (29.22) is an asymptotic expansion, accurate when $\nu^2 T$ is small. For long expirations or high volatility of volatility it degrades, and, more seriously, it can imply a *negative probability density* at low strikes, which is Section 19.5's constraint violated by an approximation rather than by the market. In the years rates spent near or below zero, that region was not a remote wing but where options actually traded.

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The responses parallel Section 29.8: a *shifted* SABR displaces F and K so the model operates away from zero, while *no-arbitrage* SABR replaces the expansion with a numerical solution guaranteed to produce a valid density. Both are in use at dealers, and which a desk employs is a material assumption rather than a technical detail.

The same holds for the whole cube: it is fitted rather than interpolated, and the fitting model is itself a source of model risk. Two desks marking the same quoted points can carry materially different values for everything in between.

Case Study 29.2 The 2022 volatility regime change

For most of the decade after the financial crisis, interest rate volatility was low and falling. Policy rates sat near zero, central banks guided expectations explicitly, and the MOVE index, which tracks implied volatility on Treasury options, spent long stretches well below its historical average.

That ended abruptly in 2022. As inflation forced central banks into the fastest tightening cycle in forty years, rate volatility rose sharply, and the MOVE index climbed through 2022 to a peak during the March 2023 banking stress at its highest level since 2008. Normal volatilities on short-dated dollar swaptions doubled or tripled from their pre-2022 levels, and longer-dated ones rose by less.

Two lessons follow for this chapter. First, positions that looked like carry trades in the low-volatility regime (selling swaptions, or the convexity trades of Section 30.3) turned out to be short volatility at prices set by a decade of calm. Second, the vol-of-vol embedded in the smile matters: a cube fitted in a quiet regime understates how much its own shape can move.

29.10 HEDGING INTEREST RATE DERIVATIVES

The Greeks of Chapter 18, the coefficients of its Taylor expansion (18.1), carry over, with rate-specific names.

Delta for a rate derivative is its sensitivity to a shift in the curve. Because a curve is not a single number, delta is not either: desks compute a **DV01** (or PV01) against a parallel one basis point shift, and a vector of **bucketed deltas** against a one basis point shift in each curve segment separately. The bucketed vector is what actually gets hedged, since a portfolio can be insensitive to a parallel shift while carrying large offsetting exposures to steepening.

Vega is sensitivity to volatility, and inherits the dimensionality of the cube: a swaption book has vega to each expiration–tenor bucket, and hedging it requires trading options at comparable points.

Gamma measures how fast delta changes as the rate moves, and so how often a hedge must be rebalanced. Backward-looking products add a second source of change. As Section 29.4 showed, a compounded-rate option's sensitivity to the rate decays through the accrual period as fixings are locked in, so its delta changes with the passage of time even when rates do not move. A hedge that is correct at the start of the period is progressively too large, so these products require rebalancing through the accrual period rather than at reset dates alone.

The caution of Section 19.11, that a delta depends on how the smile moves with the underlying, applies with extra force here. These Greeks are computed within a model that is known to be a quotation convention rather than a description of reality. A delta computed under Black and a delta computed under Bachelier differ by about $\varphi(0)\,\sigma\sqrt{T}/2$ at the money, with σ the Black volatility, so the gap grows with $\sigma\sqrt{T}$. For the at-the-money swaption of Example 29.6, Black's delta is $N(d_1) = 0.588$ per unit of annuity and Bachelier's, at the same price, is $N(0) = 0.500$: hedging the \$100 million option takes a forward swap on \$58.8 million

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under one convention and \$50.0 million under the other. Which is right depends on how volatility actually moves when the rate moves—a question the quoting model does not answer.

SUMMARY

Caps, floors, and swaptions are valued with Black's model applied to the appropriate forward quantity under the appropriate numeraire. For a caplet the forward quantity is the forward rate and the numeraire is the zero-coupon bond maturing on the payment date; for a swaption it is the forward swap rate and the annuity. These are quotation conventions rather than descriptions of rate behavior: their function is to provide an agreed, invertible map between prices and volatilities. Bond options are quoted in yield volatility and converted to price volatility by duration, and an option-adjusted spread means nothing without its volatility assumption.

A cap is a portfolio of separately settled caplets, so it is valued one period at a time. Cap–floor parity states that a cap less a floor at the same strike equals a payer swap, a model-free relationship useful as a consistency check. Market quotes use flat volatilities (a single number that reprices a whole cap), while analysis requires spot volatilities specific to each caplet, extracted by a bootstrap. The two must never be compared directly.

The move to compounded risk-free rates changed the caplet. A backward-looking rate keeps accumulating uncertainty across its accrual period, with sensitivity decaying linearly as fixings are locked in, giving total variance $\sigma^2(T_1 + \tau/3)$ rather than $\sigma^2 T_1$. Backward-looking caplets are therefore always worth more than their forward-looking equivalents, and the first caplet of a spot-starting cap, formerly a known cash flow, now carries real optionality.

Caps and swaptions on the same schedule are not equivalent: a swaption is a single option on a weighted average of the rates on which a cap holds one option each. Because a sum of positive parts is at least the positive part of the sum, the cap is worth at least as much, and the gap is governed by the correlation among forward rates, narrowing as that correlation rises toward one. Those correlations are not observable while cap and swaption prices are, so the ratio between them is the market's principal source of correlation information and the main calibration input for multi-factor term structure models.

Negative rates made the lognormal assumption not merely inaccurate but undefined, since $\ln(F_0/K)$ requires a positive forward. The market moved to normal (Bachelier) volatilities, under which an at-the-money option is worth $\sigma_N \sqrt{T/2\pi}$ per unit of numeraire, and to shifted lognormal conventions. Near the money, $\sigma_N \approx \sigma_{\mathrm{LN}} F_0$.

The swaption market is described by a volatility cube over expiration, tenor, and strike. Volatility humps along the expiration dimension and declines along the tenor dimension, since a longer swap rate averages more forwards. The cube is fitted with a parametric model, usually SABR, rather than interpolated, because arbitrary interpolation admits arbitrage; Hagan's expansion can itself imply a negative density at low strikes, which the shifted and no-arbitrage variants repair. Hedging inherits the cube's dimensionality: bucketed delta, vega by expiration and tenor, and a delta that decays through the accrual period of a backward-looking product.

FURTHER READING

Black's "The Pricing of Commodity Contracts" (1976) is the origin of the formula applied here to caps and swaptions. Bachelier's 1900 thesis is the normal model, written five years before Einstein's treatment of Brownian motion and worth reading now that rates have made the normal convention standard again. Hagan, Kumar, Lesniewski, and Woodward's "Managing Smile Risk" (2002) introduced SABR and is the source of the expansion in (29.22). Andersen and Piterbarg's three-volume *Interest Rate Modeling* is the reference to reach for when the market conventions here are not enough, and Lyashenko and Mercurio (2019) is where the $T_1 + \tau/3$ result for backward-looking rates comes from.

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PRACTICE QUESTIONS

29.1 Explain why Black's model can be applied to a caplet without any adjustment for the drift of the forward rate. Which discount factor must be used, and why that one?

29.2 A caplet covers a three-month period beginning in two years, on a notional of \$20 million. The forward rate is 3.60%, the strike is 3.75%, the volatility is 18%, and the discount factor to the payment date is 0.9280. Value the caplet, treating the rate as fixed in advance.

29.3 Repeat Question 29.2 treating the rate as compounded in arrears. By what percentage does the value increase?

29.4 State cap–floor parity and prove it from the payoffs. Explain how you would use it to check a set of broker quotes.

29.5 Explain the difference between a flat volatility and a spot volatility. Why is it wrong to compare the flat volatilities quoted for a three-year and a seven-year cap?

29.6 Explain why the first caplet of a spot-starting cap was conventionally excluded under LIBOR, and why that convention is no longer appropriate.

29.7 A five-year option on a three-year annual-pay swap has a forward swap rate of 3.80%, a strike of 3.80%, and a volatility of 22%. The annuity is 2.6540 per unit of notional. Value the payer swaption on a notional of \$50 million, and state the value of the corresponding receiver.

29.8 Show that at the money the Bachelier formula reduces to $\sigma_N \sqrt{T/2\pi}$ per unit of numeraire.

29.9 An at-the-money swaption is quoted at a lognormal volatility of 24% with a forward swap rate of 3.20%. Estimate the equivalent normal volatility, and explain why the rule of thumb works.

29.10 Explain why the volatility of a swap rate generally declines as the tenor of the underlying swap increases.

29.11 Explain, without algebra, why a cap must be worth at least as much as a swaption written on the same schedule and strike. Under what condition would the two be equal?

29.12 Eight forward rates each have a volatility of 24% and enter a swap rate with equal weight. Using (29.17), find the implied swap rate volatility when the uniform pairwise correlation is 1.0, 0.6, and 0.2. Comment on what the results imply about calibrating a model to caps alone.

FURTHER QUESTIONS

29.13 Derive the result $\mathrm{Var} = \sigma^2 (T_1 + \tau/3)$ for a backward-looking rate, stating clearly the assumption made about how sensitivity decays through the accrual period. How would the answer change if the decay were assumed to be quadratic rather than linear?

29.14 A cap consists of four quarterly caplets beginning immediately, with $\tau = 0.25$. Using a flat volatility of 20%, a strike of 4.00%, and a forward rate of 4.00% for every period, and ignoring discounting, compare the total value of the cap when the caplets are treated as forward-looking against the value when they are treated as backward-looking. Comment on where the difference is concentrated.

29.15 A trader observes that three-month expiration swaptions are quoted at 85 basis points of normal volatility while one-year expiration swaptions on the same tenor are quoted at 78. Explain what this implies about the expected path of rate uncertainty, and identify at least two reasons this need not represent an arbitrage.

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29.16 Explain why a delta computed under Black's model differs from one computed under Bachelier for the same option at the same price. Under what circumstances is the difference largest, and what additional information would you need to decide which is the better hedge ratio?

29.17 In 2022 the level of rates and the volatility of rates rose together. Explain, using the relationship between normal and lognormal volatility, why a desk quoting in normal volatilities and a desk quoting in lognormal volatilities would have reported very different percentage increases in "volatility" over that year, even though both observed the same market.

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